

Responsive Web Development Using Bootstrap

Responsive Web development was conceptualised to make websites look good across the spectrum of devices, ranging from mobiles to desktops to large screens. Bootstrap presupposes that the user has no designing knowledge whatsoever, but merely wishes to write some HTML according to Bootstrap specifications.



The Bootstrap framework is an open source collection of tools used to develop responsive (here, the word means layouts that adapt to desktops as well as for small screen devices like mobiles and tabs) websites. Bootstrap, originally named Twitter Blueprint, was developed by Mark Otto and Jacob Thornton in August 2011 to increase consistency across internal tools. It includes HTML and CSS based design templates for typography, forms, buttons, tables, image carousels and many others, as well as an optional JavaScript library. It assumes that we have no designing knowledge, and all we want to do is write some HTML as per the Bootstrap specifications. In short, Twitter Bootstrap has already written a CSS style sheet; we just have to use it.

Grid basics of Bootstrap

Bootstrap's grid system is fully responsive and allows us to divide our page into 12 columns. If we don't want to use all 12 columns individually, we can group the columns together to create wider columns using different classes.

The grid system has four classes:

1. xs (for phones)

2. sm (for tablets)
3. md (for desktops)
4. lg (for larger desktops)

These classes can also be combined together to create more dynamic and flexible layouts.

The structure of a Bootstrap grid is as follows:

```
<div class="row">
<div class="col-*"></div>
<div class="col-*"></div>
<div class="col-*"></div>
</div>
```

First, we have to define the row; then, we have to define columns according to our requirements.

Here's how we replace the first * with the class according to the screens (like xs, sm, md, lg) and the second * with column numbers between 1 to 12, but their total should not exceed 12.

For example:

```
col-md-4
```